JANUARY REPORT

Problem Analysis

7 - Jan
14 - Jan
21 - Jan

Overview

JAN	*Problem Analysis*	• **Identified disaster communication challenges in Mumbai such as floods, heavy rainfall, and transport disruptions** • **Defined a clear and focused problem statement based on real-world observations** • **Collected research papers, journals, and case studies related to disaster communication** • **Conducted stakeholder analysis including citizens, authorities, and emergency responders** • **Studied existing systems such as NDMA alerts, FEMA WEA, and social media platforms** • **Performed detailed literature review to understand gaps in current systems** • **Analyzed issues like language barriers, misinformation, lack of clarity, and absence of feedback systems**

Monthly Summary & Analysis

During this phase, the primary focus was on gaining a deep understanding of the problem domain. Various real-world challenges faced by citizens during disasters were identified, including difficulty in understanding alerts, lack of multilingual communication, and spread of misinformation. Research papers and case studies were analyzed to validate these problems and understand existing solutions. Stakeholder analysis helped in identifying user needs and expectations from a disaster communication system. This phase played a crucial role in building a strong conceptual foundation and clearly defining the interdisciplinary nature of the project by connecting design thinking, software engineering, and database concepts.

FEBRUARY REPORT

Solution Design

4 - Feb
11 - Feb
18 - Feb
25 - Feb

Overview

Feb	*Solution Design*	• **Generated multiple innovative solution ideas to address identified problems** • **Evaluated and selected the most feasible solution (CrisisClarity platform)** • **Defined clear objectives, scope, and expected outcomes of the system** • **Identified key features such as multilingual alerts, AI chatbot, feedback system, and location-based filtering** • **Conducted feasibility analysis (technical, economic, and operational)** • **Designed high-level system architecture and workflow** • **Selected appropriate technologies including Flutter, FastAPI, Firebase, and AI APIs** • **Planned integration of external services such as RSS feeds and Telegram bot**

Monthly Summary & Analysis

This phase focused on converting the problem into a structured and feasible solution. Multiple ideas were explored using design thinking approaches, and the most effective solution was selected based on practicality and impact. The system design was carefully planned to ensure scalability, usability, and efficiency. Key decisions such as technology stack selection and feature inclusion were made based on real-world requirements and feasibility. The integration of AI, multilingual capabilities, and feedback mechanisms highlights innovation and strong application of interdisciplinary knowledge. This phase ensured that the solution is not only technically sound but also user-centric and impactful.

MARCH REPORT

System Development

3 - Mar
11 - Mar
18 - Mar
25 - Mar

Overview

Mar	*System Development*	• Prepared System Requirements Specification (SRS) document • Designed detailed system architecture and workflow diagrams • Created UI wireframes and initial prototypes for the application • Started frontend development using Flutter • Developed backend APIs using FastAPI • Implemented basic features such as alert display and user interaction • Initiated AI integration for chatbot and alert processing • Connected Firebase for database and real-time data handling • Began testing of individual modules and basic functionalities

Monthly Summary & Analysis

In this phase, the project transitioned from planning to actual implementation. Detailed system design and documentation were completed to guide development. Frontend and backend development began simultaneously, ensuring proper integration between components. Core functionalities such as alert display, API communication, and database connectivity were implemented. Initial AI integration marked the beginning of intelligent system behavior. This phase demonstrated strong technical skills and interdisciplinary application, combining concepts from software engineering, mobile development, databases, and AI. It laid a solid foundation for further development, testing, and final system deployment.